Homework 2: Nico Boyle, Owen Moorhus, Asher Olgeirson

1. Let k be an integer.
   Claim: If $5k + 3$ is even, then k is odd.
   Prove the claim by first converting to contrapositive.

   $Proof$:
   By contrapositive, assume k is an even integer
   $k = 2n$ where n is an integer
   $5k + 3 = 5(2n) + 3$
   $= 10n + 3$
   $= 10n + 2 + 1$
   $= 2(5n + 1) + 1$
   $= (2j) + 1$ where j is an integer
   $= 5k + 3 = 2j + 1$ which is odd by definition, thus proving k is odd by contrapositive
   ■

2. Suppose that x and y are positive real numbers. Prove by contradiction that (for all such x, y) $x + y \geqslant 2\sqrt{xy}$

$Proof:$
by contradiction, assume $x + y < 2\sqrt{xy}$
$x^2 + 2xy + y^2 < 4xy$
$x^2 - 2xy + y^2 < 0$
$(x - y)^2 < 0$
This is a contradiction since the square of a real number is always non-negative.
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3. Prove the following, by smallest counterexample.

   For all $n \geqslant 1$, $\sum\limits_{k=1}^{n} k(k+1) = \frac{n(n+1)(n+2)}{3}$